# Research-Grade Correction and Validation of Homogeneous Elasticity in PTO/STO Superlattices

## Physical conventions, implementation, numerical validation, and remaining model boundaries

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### Purpose and main conclusion

This report documents the research-grade correction of the **homogeneous-elasticity module** used by the OpenFerro PTO/STO examples, so that the implementation can be reviewed, reproduced, and extended. Before the correction, the model included only the PTO homogeneous elastic energy, set the STO homogeneous strain–dipole coupling to zero, and did not explicitly transform the shared-cell strain to the bulk reference lattice of each constituent. After the correction:

- PTO and STO use their own elastic constants, strain–dipole parameters, and actual numbers of unit cells;
- deformation of the shared coherent lattice is transformed to strain relative to the reference lattice of each material;
- both materials contribute energy and autodifferentiated force to the same global strain field;
- unit tests, finite differences, constrained relaxation, and minimal build/run checks for both examples have passed.

The model has therefore advanced from a "PTO-only homogeneous-elastic approximation" to a **two-material homogeneous-elastic model with coherent PTO/STO lattice mismatch**. It is still not a complete elastic model of a heterostructure because local acoustic displacements, inhomogeneous elasticity, and layer-resolved strain relaxation remain absent.

# Contents

**1 Scope of the correction and defects in the previous model** **3**
1.1 Simulated systems . . . 3
1.2 Three substantive problems in the previous implementation . . . 3

**2 Corrected physical conventions** **3**
2.1 Engineering Voigt strain . . . 3
2.2 Two-material homogeneous elastic energy . . . 4
2.3 Homogeneous strain–dipole coupling . . . 4

**3 Material parameters and provenance** **4**

**4 Implementation and backward compatibility** **5**
4.1 Material-relative strain mapping . . . 5
4.2 System API extension . . . 5
4.3 Superlattice strain–dipole engine . . . 6
4.4 Registration in the two PTO/STO examples . . . 6

**5 Numerical diagnostics: what changed?** **6**
5.1 Material-relative strain . . . 6
5.2 Quantitative homogeneous-elastic comparison . . . 6

**6 Validation protocol and results** **7**
6.1 Test matrix . . . 7
6.2 Executed checks . . . 7
6.3 Computational environment . . . 8

**7 Remaining model limitations** **8**

**8 Future work and review checklist** **9**
8.1 Recommended development order . . . 9
8.2 Minimum checks after each update . . . 9

**A Index of modified files** **9**

**B Reproduction commands** **9**

# 1 Scope of the correction and defects in the previous model

## 1.1 Simulated systems

The two corrected examples are:

- `examples/04.PTOSTO_superlattice/npt.py`: single-device, field-driven PTO/STO dynamics;
- `examples/04a.PTOSTO_superlattice_large/npt.py`: cooling calculations for larger, potentially multi-device systems.

Both construct a simple-cubic computational grid using the PTO lattice constant $a_c = a_{PTO} = 3.950$ Å, with stacking along $z$. The initial in-plane strain is fixed at

$$\eta_{xx} = \eta_{yy} = -0.012, \tag{1}$$

so the actual in-plane lattice constant is

$$a_{\parallel} = 3.950(1 - 0.012) = 3.9026 \text{ Å}, \tag{2}$$

which is very close to the $3.901$ Å reference value in the STO parameter record.

## 1.2 Three substantive problems in the previous implementation

1. **The STO homogeneous elastic energy was omitted.** The old code registered only one PTO elastic term, with a cell count covering only the PTO layers. The restoring force on the shared global strain was therefore underestimated.
2. **The STO homogeneous strain–dipole coupling was disabled by hand.** The STO $(B_{1xx}, B_{1yy}, B_{4yz})$ coefficients were entered as $(0, 0, 0)$ in the examples, preventing the STO polar soft mode from responding to homogeneous strain.
3. **The reference lattices were conflated.** Directly inserting the same numerical value $\eta = -1.2\%$ into both the PTO and STO bulk elastic energies would assert that both materials are compressed by $1.2\%$ relative to their own bulk lattices. In the actual calculation, however, the $3.9026$ Å in-plane lattice is nearly unstrained for STO.

**Why this is not a small parameter patch**

The problem affects the equilibrium global strain, elastic restoring force, polarization–strain feedback, and hence the stability of domains and topological textures. Adding an STO elastic term without resolving the reference-lattice convention would place STO in the wrong prestrained state. Correcting the mismatch while leaving the STO strain–dipole coefficients at zero would likewise be incomplete.

# 2 Corrected physical conventions

## 2.1 Engineering Voigt strain

OpenFerro stores global strain in engineering Voigt order,

$$\boldsymbol{\eta} = (\eta_{xx}, \eta_{yy}, \eta_{zz}, 2\eta_{yz}, 2\eta_{xz}, 2\eta_{xy}). \tag{3}$$

The shared computational lattice is referenced to $a_c$. For a cubic material $m$ whose bulk reference lattice constant is $a_m$, define

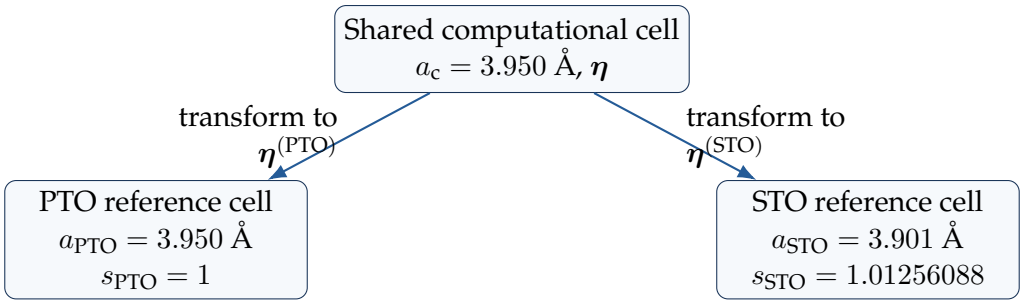
$$s_m = \frac{a_c}{a_m}. \tag{4}$$

The same actual deformation, expressed relative to material $m$, is

$$\eta_{ii}^{(m)} = s_m(1+\eta_{ii}) - 1, \qquad i = x, y, z, \tag{5}$$

$$\eta_{ij}^{(m)} = s_m \eta_{ij}, \qquad i \neq j. \tag{6}$$

This re-expresses the shared deformation gradient relative to the material's own cubic reference cell. The present formula is appropriate for the cubic reference lattices and engineering small-strain energy expansion used here.




### 2.2 Two-material homogeneous elastic energy

For material $m$, the cubic homogeneous elastic energy is

$$E_{\text{elas}}^{(m)} = N_m \left[ \frac{B_{11}^{(m)}}{2} \sum_{a=1}^{3} \left(\eta_a^{(m)}\right)^2 + B_{12}^{(m)} (\eta_1^{(m)}\eta_2^{(m)} + \eta_2^{(m)}\eta_3^{(m)} + \eta_3^{(m)}\eta_1^{(m)}) + \frac{B_{44}^{(m)}}{2} \sum_{a=4}^{6} \left(\eta_a^{(m)}\right)^2 \right], \tag{7}$$

and the total homogeneous elastic energy is

$$E_{\text{elas}} = E_{\text{elas}}^{(\text{PTO})} + E_{\text{elas}}^{(\text{STO})}. \tag{8}$$

Here,

$$N_{\text{PTO}} = L_x L_y m_{\text{PTO}}, \qquad N_{\text{STO}} = L_x L_y n_{\text{STO}}. \tag{9}$$

Both terms act on the same global strain field, so the total autodifferentiated strain force is the sum of the two material contributions.

### 2.3 Homogeneous strain–dipole coupling

The local field in the PTO/STO examples stores the physical dipole moment $\mathbf{p}$ in $e$Å, whereas the local soft mode in the parameter records is the displacement $\mathbf{u}$. Because $\mathbf{p} = Z^*\mathbf{u}$, the strain–dipole coefficients used in the examples are divided by $(Z^*)^2$.

Each material region uses its own $\boldsymbol{\eta}^{(m)}$ and normalized coupling coefficients. The interface layers retain the arithmetic-average convention of the original model,

$$\mathbf{B}_{\text{I}} = \frac{\mathbf{B}_{\text{PTO}} + \mathbf{B}_{\text{STO}}}{2}, \tag{10}$$

while the two interface-cell faces use the material-relative strain of the material occupying each face. This preserves the existing interface-parameter convention without conflating the PTO and STO reference lattices.

## 3 Material parameters and provenance

The material records and literature sources are:

- PTO: Paul *et al.*, *Phys. Rev. B* **95**, 054111 (2017), DOI: 10.1103/PhysRevB.95.054111; the local record is labeled as a SCAN effective Hamiltonian.

| Parameter | PTO | STO | Description |
|---|---|---|---|
| Reference lattice $a$ (Å) | 3.950 | 3.901 | Values in the local material records |
| $B_{11}$ (eV/cell) | 131.76 | 131.33 | Cubic elastic coefficient |
| $B_{12}$ (eV/cell) | 52.32 | 36.26 | Cubic elastic coefficient |
| $B_{44}$ (eV/cell) | 41.85 | 41.30 | Engineering-shear convention |
| $B_{1xx}$ (eV/Å$^2$) | $-66.37$ | $-102.09$ | Soft-mode–strain coupling |
| $B_{1yy}$ (eV/Å$^2$) | $-2.99$ | 0.5299 | Soft-mode–strain coupling |
| $B_{4yz}$ (eV/Å$^2$) | $-23.25$ | $-15.494$ | Shear coupling |
| $Z^*$ ($e$) | 9.37 | 9.28 | Born effective charge |
| $\epsilon_\infty$ | 7.95 | 6.46 | Current Ewald term still uses PTO value throughout |

Table 1: Material parameters read by the corrected implementation.

- STO: Nishimatsu *et al.*, *J. Phys. Soc. Jpn.* **85**, 114714 (2016), DOI: 10.7566/JPSJ.85.114714; the local record is labeled as a Wu–Cohen GGA effective Hamiltonian.

  The normalized coupling coefficients used with the physical dipole field are

$$\widetilde{\mathbf{B}}_{\mathrm{PTO}} = (-0.75594924, -0.03405587, -0.26481573), \tag{11}$$

$$\widetilde{\mathbf{B}}_{\mathrm{STO}} = (-1.18546095, \quad 0.00615316, -0.17991509). \tag{12}$$

**Parameter-consistency risk**

The PTO and STO parameters come from different exchange–correlation functionals and separate effective-Hamiltonian fits. The present correction ensures that the local records are used consistently, but it cannot eliminate the systematic error introduced by combining unlike parameter sets. Quantitative publication work should seek PTO/STO parameters at a common theoretical level or include parameter-set differences in a sensitivity analysis.

# 4 Implementation and backward compatibility

## 4.1 Material-relative strain mapping

The following pure-JAX function was added to `openferro/engine/elastic.py`:

```
def material_homogeneous_strain(global_strain, lattice_scale=1.0):
    scale = jnp.asarray(lattice_scale, dtype=global_strain.dtype)
    normal = scale * (1.0 + global_strain[:3]) - 1.0
    shear = scale * global_strain[3:]
    return jnp.concatenate((normal, shear))
```

`homo_elastic_energy` accepts both the original four-parameter array

$$(B_{11}, B_{12}, B_{44}, N) \tag{13}$$

and the new five-parameter array

$$(B_{11}, B_{12}, B_{44}, N, s_m). \tag{14}$$

Older models therefore retain the default $s_m = 1$ and keep their previous numerical meaning.

## 4.2 System API extension

The interface in `openferro/system.py` now includes

```
add_homo_elastic_interaction(..., n_cells=None, lattice_scale=1.0)
```

and validates `n_cells` as a positive integer and `lattice_scale` as a finite positive scalar. Each constituent can therefore specify its own unit-cell count and reference-lattice ratio explicitly.

### 4.3 Superlattice strain–dipole engine

The factory function in `openferro/engine/ferroelectric_superlatt.py` now accepts

```
get_homo_strain_dipole_interaction_on_AmBnLattice(
    lattice, m, n, lattice_scales=(scale_A, scale_B)
)
```

The interiors of both materials and the two interface-cell faces are summed separately. PTO and STO use different material-relative strains, while the interface coupling coefficients remain arithmetic averages.

### 4.4 Registration in the two PTO/STO examples

Both examples now register

```
homo_elastic_pto  # PTO constants, PTO cell count, scale=1
homo_elastic_sto  # STO constants, STO cell count, scale=3.950/3.901
homo_strain_dipole  # nonzero PTO and STO coupling coefficients
```

The same convention is documented in:

- the root `README.md`;
- the README under `examples/04.PTOSTO_superlattice/`;
- `docs/source/scientific_conventions.rst`.

## 5 Numerical diagnostics: what changed?

### 5.1 Material-relative strain

For the shared initial strain used by the examples,

$$\boldsymbol{\eta} = (-0.012, -0.012, 0.051, 0, 0, 0), \tag{15}$$

the PTO- and STO-relative strains are

$$\boldsymbol{\eta}^{(\mathrm{PTO})} = (-0.012, -0.012, 0.051, 0, 0, 0), \tag{16}$$

$$\boldsymbol{\eta}^{(\mathrm{STO})} = (0.00041015, 0.00041015, 0.06420149, 0, 0, 0). \tag{17}$$

Thus, the shared in-plane value of $-1.2\%$ means $-1.2\%$ compression for PTO but only about $+0.041\%$ tension for STO. Recovering this distinction is the central purpose of the reference-lattice correction.

### 5.2 Quantitative homogeneous-elastic comparison

The following diagnostic uses 96 PTO cells and 96 STO cells and retains only homogeneous elastic energy. Polarization–strain coupling, Ewald energy, temperature, and external pressure are excluded.

STO favors a material-relative $z$ strain near zero. Because the shared reference cell uses the larger PTO lattice, this corresponds to a shared $\eta_{zz}$ near $3.901/3.950 - 1 \approx -1.24\%$. In-plane-compressed PTO instead favors positive out-of-plane extension. The two-material minimum lies between these preferences.

**Do not interpret $-0.167\%$ as the equilibrium out-of-plane strain of the full simulation**

The table is a purely elastic analytic test used to audit the implementation. A full PTO/STO trajectory also contains nonzero local dipoles, homogeneous strain–dipole coupling, long-range dipole interactions, thermal fluctuations, and possibly a pressure term. In particular, ferroelectric polarization in PTO strongly favors tetragonal elongation. The equilibrium $\eta_{zz}$ of the full model must therefore be obtained from full-Hamiltonian dynamics or minimization.

| Diagnostic quantity | PTO | STO | Total or interpretation |
|---|---|---|---|
| Energy at initial $\eta_{zz} = 0.051$ (eV) | 12.846885 | 26.169393 | 39.016278 |
| Initial generalized $z$ force (eV) | $-524.55168$ | $-822.49029$ | $-1347.04197$ |
| Energy at corrected purely elastic minimum (eV) | 2.763604 | 0.778641 | 3.542245 |
| Optimal $\eta_{zz}$ in old PTO-only model | 0.0095301 | | PTO Poisson response only |
| Optimal $\eta_{zz}$ in corrected two-material model | $-0.0016695$ | | PTO/STO competition |

Table 2: Purely homogeneous-elastic diagnostic. The generalized force is $-\partial E/\partial \eta_{zz}$.

# 6 Validation protocol and results

## 6.1 Test matrix

| Test | Failure mode excluded | Acceptance criterion | Result |
|---|---|---|---|
| Material-strain mapping | STO incorrectly compressed by another 1.2%; incorrect shear scaling | Componentwise agreement with analytic formula; STO in-plane strain below $5 \times 10^{-4}$ | Pass |
| Linear cell-count scaling | Missing or duplicated layer count | $E(N_1) + E(N_2) = E(N_1 + N_2)$ | Pass |
| Homogeneous-material limit | Artificial energy change when one material is divided into layers | Sum of two identical-material regions equals one bulk region | Pass |
| Autodiff versus finite difference | Error in JAX gradient, material mapping, or parameter order | Forces for all six strain components agree within tolerance | Pass |
| JIT versus eager | Closure or static branch changes numerical results after compilation | Compiled and eager energies agree | Pass |
| Two-variable strain–dipole force | Strain force is correct but dipole force is not | Separate finite differences for strain and four-dimensional dipole array | Pass |
| Constrained $\eta_{zz}$ relaxation | Incorrect minimum or curvature with fixed in-plane strain | Gradient vanishes at minimum and either perturbation raises energy | Pass |
| Example parameter registration | STO term remains zero or layer count is wrong | Both elastic terms exist and all three STO coupling components are nonzero | Pass |
| Example 04 minimal trajectory | System builds but cannot integrate | One zero-field and one finite-field step for $4 \times 4 \times (2+2)$ complete and write output | Pass |
| Example 04a full-force build | Large-system script differs from example 04 | Dipole and strain forces are finite for a minimal cell | Pass |

## 6.2 Executed checks

The core commands were

```
JAX_PLATFORMS=cpu .venv/bin/python -m pytest \
  tests/unit_tests/test_elastic.py \
  tests/unit_tests/test_ferroelectric.py -q
# 20 passed
```

```
JAX_PLATFORMS=cpu .venv/bin/python -m pytest \
  tests/unit_tests/test_examples.py -q -k 'PTO_STO or ptosto'
# 2 passed, 6 deselected
```

A further compatibility check covered `test_system.py`, `test_integrator_md.py`, and related examples, with 30 tests passing. The full non-GPU unit-test audit produced 204 passes and 6 skips. The remaining one failure and seven initialization errors came from missing optional `matplotlib` support in nanoparticle plotting analysis and were unrelated to the PTO/STO elastic correction. Network access was unavailable when installation of that optional dependency was attempted.

### 6.3 Computational environment

- Python 3.13.12, JAX 0.11.2, and pytest 9.2.0;
- `JAX_PLATFORMS=cpu` was used for this audit;
- NVML/CUPTI could not be initialized in the audit session, so GPU-backend regression was not completed;
- no production-size, long-time statistical trajectory was included. These checks establish implementation correctness and short-run executability, not physical convergence.

## 7 Remaining model limitations

1. **No local or inhomogeneous elasticity.** Only one shared six-component global strain is present. Every PTO and STO layer is forced to share the same out-of-plane lattice extension, so the model cannot form layer-resolved $c(z)$, interface strain gradients, or local shear textures.
2. **No local strain–dipole coupling.** A spatially varying polarization cannot further release elastic energy through local acoustic displacements. This may alter vortex cores, domain walls, and the quantitative stability window of topological textures.
3. **Arithmetic-average interface parameters.** This simple convention comes from the original model rather than a dedicated first-principles fit of a PTO/STO interface. Future studies should scan the interface scaling factor as an uncertain parameter.
4. **Uniform dielectric screening.** The Ewald term still uses the PTO value $\epsilon_\infty = 7.95$ throughout and contains no spatially varying dielectric response.
5. **Inconsistent theoretical levels across parameter sets.** The PTO record uses SCAN, whereas the STO record uses WC-GGA.
6. **No explicit oxygen-octahedron rotations in STO.** At low temperature or strong strain, antiferrodistortive modes may couple to polarization and topological textures.

**Conclusions supported by the correction**

One may state that the current examples 04 and 04a include self-consistent energy and force contributions from both PTO and STO to a shared homogeneous strain and correctly represent the coherent mismatch between the two bulk reference lattices. The model is suitable for preliminary studies of three-dimensionally periodic topological dynamics in which homogeneous elasticity participates, and for model-ablation studies.
One may not state that the model fully represents the elastic field of a real PTO/STO superlattice, or that its equilibrium strain, phase boundaries, and topological stability regions have quantitative predictive accuracy.

# 8 Future work and review checklist

## 8.1 Recommended development order

1. Freeze this corrected homogeneous-elasticity version. Establish regression records for short trajectories and for individual energy components.
2. Add material-resolved local acoustic displacement fields, inhomogeneous elasticity, and local strain–dipole coupling.
3. Compare homogeneous-only and full local-elasticity models using identical initial states, sizes, and temperatures.
4. Scan interface coupling, the uniform dielectric constant, and differences between parameter sets.
5. Only then begin production calculations of Lyapunov exponents, topological-charge time series, and intrinsic chaos over multiple superlattice periods.

## 8.2 Minimum checks after each update

- `git diff --check` reports no whitespace errors;
- material mismatch, cell-count scaling, finite-difference force, and constrained-relaxation tests continue to pass;
- PTO/STO elastic energies, strain–dipole energy, and total strain force are finite when recorded separately;
- total and component energies for a fixed initial state are recorded as a numerical baseline;
- when a GPU backend is available, JIT/GPU parity and at least one short NVT/NVE trajectory are added;
- every material-parameter update records its source, units, reference lattice, and Born-charge normalization.

# A Index of modified files

| File | Role in this correction |
|---|---|
| `openferro/engine/elastic.py` | Material-relative strain mapping and four/five-parameter-compatible homogeneous elastic energy |
| `openferro/system.py` | Public `n_cells` and `lattice_scale` arguments with validation |
| `openferro/engine/ferroelectric_superlatt.py` | Homogeneous strain–dipole coupling using separate PTO and STO reference strains |
| `examples/04.PTOSTO_superlattice/npt.py` | Two-material homogeneous elasticity and nonzero STO coupling |
| `examples/04a.PTOSTO_superlattice_large/npt.py` | Same research-grade correction as example 04 |
| `tests/unit_tests/test_elastic.py` | Mismatch, scaling, homogeneous limit, finite differences, and constrained relaxation |
| `tests/unit_tests/test_ferroelectric.py` | Two-variable finite differences and JIT checks for superlattice strain–dipole coupling |
| `tests/unit_tests/test_examples.py` | Actual example registration and minimal trajectory tests |
| `README.md` and documentation | Removal of obsolete “missing STO elasticity” statement and documentation of new convention and remaining limitations |

# B Reproduction commands

Run the following from the `OpenFerro-test` repository root:

```
# Core scientific tests
JAX_PLATFORMS=cpu .venv/bin/python -m pytest \
  tests/unit_tests/test_elastic.py \
  tests/unit_tests/test_ferroelectric.py -q

# PTO/STO example registration and smoke trajectory
JAX_PLATFORMS=cpu .venv/bin/python -m pytest \
  tests/unit_tests/test_examples.py -q -k 'PTO_STO or ptosto'

# Syntax and whitespace checks
python -m compileall -q openferro \
  examples/04.PTOSTO_superlattice/npt.py \
  examples/04a.PTOSTO_superlattice_large/npt.py
git diff --check
```

**Version record**

This report describes the implementation committed as `41b8336`. Any later refactor or material-parameter change should record a new commit hash and regenerate the numerical baselines. The PDF captures the state at generation time and does not replace Git history.